

## Part B

### Q5

1. D
2. D
3. D
4. D
5. C
6. C
7. C
8. C
9. A
10. A
11. A
12. A
13. B
14. B
15. B

### Q6

Diagram the LR(1) states for the given grammar , and construct the LR(1) parsing table. Indicate clearly any conflicts.

0.  $S \rightarrow G \$$

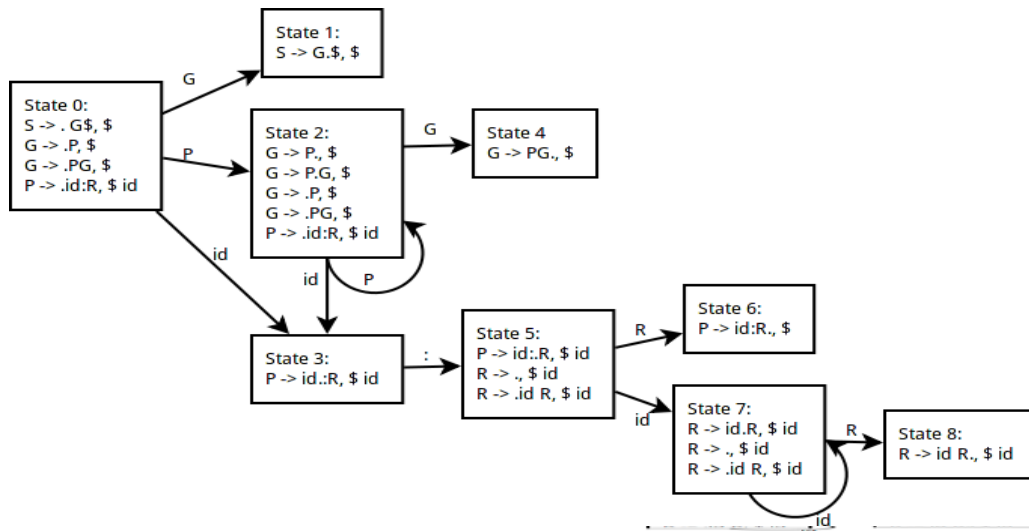
1.  $G \rightarrow P$

2.  $G \rightarrow PG$

3.  $P \rightarrow id : R$

4.  $R \rightarrow$

5.  $R \rightarrow id R$



|   | :  | $\epsilon$ | id      | \$ | S | G | P | R |
|---|----|------------|---------|----|---|---|---|---|
| 0 |    |            | s3      |    |   | 1 | 2 |   |
| 1 |    |            |         | a  |   |   |   |   |
| 2 |    |            | s3      | r1 |   | 4 | 2 |   |
| 3 | s5 |            |         |    |   |   |   |   |
| 4 |    |            |         | r2 |   |   |   |   |
| 5 |    |            | r4   s7 | r4 |   |   |   | 6 |
| 6 |    |            | r3      | r3 |   |   |   |   |
| 7 |    |            | r4   s7 | r4 |   |   |   | 8 |
| 8 |    |            | r5      | r5 |   |   |   |   |

(5,id)(7,id), id:id:

| Stack | Input          | Actio                                                                                                                                                                                               |
|-------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <u>id:id</u> : | s3                                                                                                                                                                                                  |
| id    | :id:           | s5                                                                                                                                                                                                  |
| id :  | id:            | either r4 or s7 is ok; in the first case, the program is in the form of "PG", where P is "id:" and G is "id:~"; in the later case ,the program is in the form of "P", where P is " <u>id:id</u> :". |

(c ) Not LR(1) DUE TO SHIFT REDUCE CONFLICTS AT STATE 5 AND 7.